University of St.Gallen

PROF. DR. ALEXANDER BRAUN
ASSOCIATE PROFESSOR OF INSURANCE AND CAPITAL MARKETS

Stochastic Modelling in Finance and Insurance

Problem Set 1

March 2, 2026

Instructions

Please submit your answers as a **text-based file** (PDF or Word) containing all required deliverables, such as numerical results, figures, and brief discussions. Add your properly annotated and executable **R code**. You are encouraged to use code snippets from the course and **Generative AI** for tutoring, coding, and writing. However, please remember that over-reliance on AI may compromise your learning effect. Finally, all work must be your own; copying from others is strictly prohibited.

Exercise 1: Simulating a GBM

Assume a stock with a current price of \$100 follows a geometric Brownian motion (GBM).

$$\mathrm{d}\tilde{S}_t = \mu \tilde{S}_t \, \mathrm{d}t + \sigma \tilde{S}_t \, \mathrm{d}\widetilde{W}_t,$$

with instantaneous arithmetic mean return rate $\mu = 8\%$ p.a. and instantaneous volatility of returns $\sigma = 20\%$.

a) Fix the random number generator (R command: `set.seed(13)`) and simulate <u>one path</u> of discrete returns and the corresponding price path over $T = 1$ year, using daily time steps $\Delta t = 1/250$ (trading days) and the Euler discretization:

$$\tilde{R}_t = \frac{\tilde{S}_t - S_{t-1}}{S_{t-1}} = \mu \, \Delta t + \sigma \, \Delta \widetilde{W}_t.$$

Report: A plot with your results.

b) Use your path from a) to compute the sample mean of the daily arithmetic returns:

$$\bar{R} = \frac{1}{s} \sum_{t=1}^{s} r_t$$

and annualize it via $\hat{\mu} = 250 \cdot \bar{R}$.

Report: $\hat{\mu}$ and briefly explain why it differs from μ for a single simulated path.

c) Now simulate $N = 10'000$ independent GBM return paths (same parameters), compute the average return across all paths and annualize it.

$$\hat{\mu}_{MC} = 250 \cdot \frac{1}{N \cdot s} \sum_{i=1}^{N} \sum_{t=t}^{s} r_{i,t}$$

Use `set.seed(13)` for this simulation as well, so results are reproducible.

Report: $\hat{\mu}_{MC}$ and a short comment comparing it to $\hat{\mu}$ from b) and to the true μ. What do you observe and why?

d) Recall that the exact solution for the SDE of the GBM is:

$$\tilde{S}_t = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma \widetilde{W}_t\right).$$

Use this solution to simulate 10'000 realizations of the stock price $\tilde{S}_T$ in $T = 3$ years.

Report: A histogram of your realizations. A brief comment on its shape. A comparison of the sample average of your realizations to the theoretical expectation $\mathbb{E}[\tilde{S}_T] = S_0 \exp(\mu T)$. What do you observe and why?

Exercise 2: Double digital option

The stock follows a GBM with the same parameters as in Exercise 1. Switch to the risk-neutral measure $\mathbb{Q}$, under which the stock dynamics satisfy:

$$\mathrm{d}\tilde{S}_t = r\tilde{S}_t \, \mathrm{d}t + \sigma \tilde{S}_t \, \mathrm{d}\widetilde{W}_t^{\mathbb{Q}},$$

with continuously compounded risk-free rate $r = 1.5\%$.

Consider a double digital option with maturity $T = 2$ years and payoff:

$$\widetilde{D}_T = H \cdot \mathbf{1}\{K_1 \leq \tilde{S}_T \leq K_2\},$$

where $K_1 = \$105$, $K_2 = \$115$, $H = \$20'000$.

Intuitively, this exotic option pays a lump sum of deterministic size H if the price of the underlying stock at maturity T is between the lower strike price K_1 and the upper strike price K_2. Otherwise, the option's payoff is 0.

Use a simulation to price this double digital option.

[**Hint**: you may use the exact GBM solution under $\mathbb{Q}$ instead of simulating full paths!]

Report: The option price: $V_0 = \exp(-rT) \cdot \mathbb{E}^{\mathbb{Q}}[\widetilde{D}_T]$.

Exercise 3: Guarantee Product

Consider a structured product with an annual guaranteed interest rate g and a participation in the yearly returns of a stock index $\tilde{I}_t$ at rate α. Given an investment of π_0, the product's payoff $\tilde{P}_T$ at maturity T is defined as follows:

$$\tilde{P}_T = \pi_0 \prod_{t=1}^{T} \left(1 + \tilde{R}_t^p\right)$$

with

$$\tilde{R}_t^p = \max\left(g, \alpha\left(\frac{\tilde{I}_t - \tilde{I}_{t-1}}{\tilde{I}_{t-1}}\right)\right).$$

Intuitively, each year, the invested capital is compounded with the larger of two rates: The guaranteed interest rate g or the index return in the respective year times the participation rate α. Assume that the evolution of the index $\mathrm{d}\tilde{I}_t$ can be modeled with a GBM.

- Investment: $\pi_0 = \$100'000$
- Current index value: $I_0 = 1'000$ points
- Drift and volatility: $\mu = 5\%$ p.a. and $\sigma = 13\%$ p.a.
- Risk-free interest rate: $r = 1\%$ p.a.
- Annual guaranteed interest rate: $g = 2\%$ p.a.
- Participation rate: $\alpha = 30\%$
- Maturity: $T = 20$ years

a) Simulate 100'000 paths of the index $\tilde{I}_t$ with discrete annual time steps. Use exact closed-form transitions instead of Euler discretization.
Report: A histogram of the index values at time T ($\tilde{I}_T$).

b) For each path of $\tilde{I}_t$, compute the corresponding path of $\tilde{r}_t^p$.
Use your simulation results to determine the realizations of $\tilde{P}_T$.
Report: A histogram of $\tilde{P}_T$.

c) **Report:** The present value of the expected payoff $\mathbb{E}^{\mathbb{Q}}(\tilde{P}_T)$, i.e., the price of the product. An economic explanation why the price differs from the initial investment.
[**Hint**: pricing is done under the risk-neutral measure $\mathbb{Q}$!]

Exercise 4: GenAI in Financial Engineering

Use GenAI (e.g., ChatGPT, Gemini) to create an innovative derivative product (beyond standard vanilla options) on equity risk. Pick your own parameter values for a GBM and price the product under the risk-neutral measure $\mathbb{Q}$.

Report: Your prompts, an intuitive description of the product, its formal payoff function, and its price.